

5G-based drone detection service for airports and critical infrastructure, leveraging network capabilities for airspace security.

Transportation and Logistics × Drone detection and airspace protection × Private 5G / LTE · OS088 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
80 is the world moving	54 can we play, can we win	HIGH 1 named in the evidence	€2.4bn partial evidence · per year	NEXT regulation adopted or propose...	L1 18 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

We offer a 5G-based drone detection service for airports and critical infrastructure, using our private mobile network expertise to detect and alert on unauthorized drones. This opportunity exists now because drone threats to airports are rising, and regulators are pushing for better detection, while network-based detection is becoming a proven approach.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€3.8bn €831m – €22.2bn	€2.4bn €518m – €13.8bn	€28.4m €6.2m – €166m	partial
Observed: tendered contract value, annualised	€115m	€115m	€1.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Private 5G / LTE	32.5 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Private 5G / LTE: No private-network series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

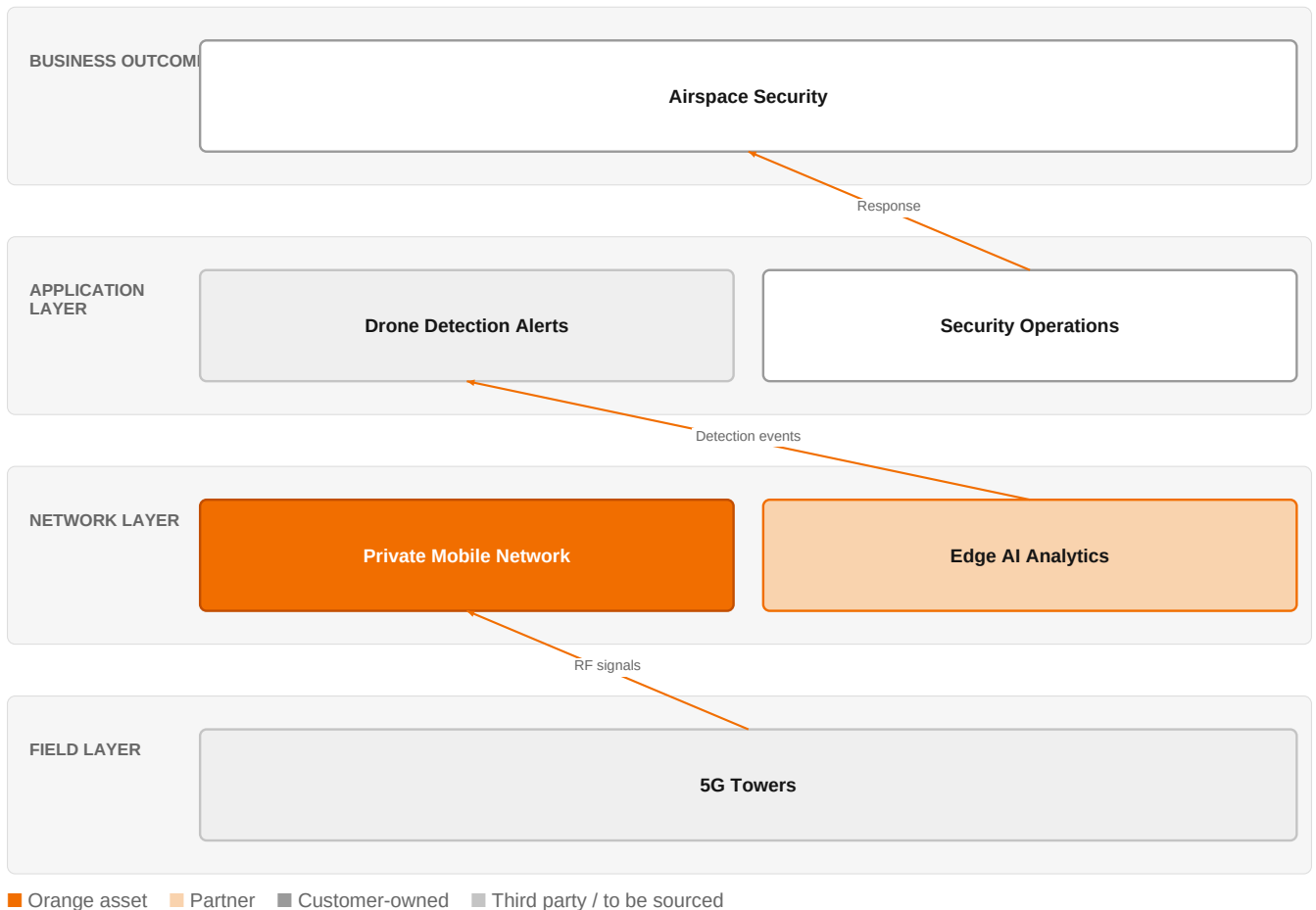
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

5G Drone Detection Service Architecture



This diagram shows how Orange's private network and drone detection service integrate with customer security operations to protect airspace. Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

Who buys, and why now

The head of airport operations or chief security officer signs, but the pain is felt by air traffic control and security teams who must detect drones that can disrupt flights. The trigger is a drone incursion that causes flight delays or a security scare, or a regulatory mandate to improve detection. They need a reliable, always-on detection system that doesn't rely on manual observation.

Why it is hot now — every claim bound to a dated source

- Lockheed Martin is preparing a 5G drone detection service using Verizon's network, indicating a market for network-based drone detection. [SIG-AE409C225269]
- Drone threats are forcing a new look at US airport security, highlighting the need for advanced detection systems. [SIG-69003C1E5931]
- Lockheed Martin plans to offer a 5G drone detection service using Verizon's network. [SIG-AE409C225269]
- Germany launched a drone testing hub after a severe airport security breach. [SIG-40242B367CE4]
- Reports highlight drone threats to airports and gaps in operator management. [SIG-22C234101FAF, SIG-A13AF8F22F6B, SIG-D2538605FC42]
- Drone threats have prompted a new look at US airport security. [SIG-69003C1E5931]
- Lockheed Martin is preparing a 5G drone detection service using Verizon's network and technology from Nvidia, ODC, Keysight and Astris AI. [SIG-AE409C225269]
- Germany launched a drone testing hub following a severe airport security breach. [SIG-40242B367CE4]

What Orange would deliver, and why it can win

What Orange would deliver

We would assemble a drone detection service using our Private mobile networks as the sensing layer, combined with our Orange Drone Guardian offer for detection and alerting. Our Cybersecurity experts and Defense and security experts would ensure the service is secure and tailored to the customer's environment. The engagement would start with a site assessment, then deploy private network coverage and integrate the drone detection software, followed by a pilot and full rollout.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Drone Guardian	offer	L1	offer addresses the use case but not this technology	unconfirmed
Private mobile networks	offer	L1	offer provides the technology in a matching domain	unconfirmed
Samsung	partner	L2	partner provides the required technology	unconfirmed
Huawei	partner	L2	partner provides the required technology	unconfirmed
Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader)	analyst position	SUP	analyst recognition covers this technology or use case	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

HIGH competitive intensity (score 9.0; 1 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
Verizon Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions, Cybersecurity	evidenced	SIG-AE409C225269, SIG-35ECED6AF5BB
BT Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions, Cybersecurity	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions, Cybersecurity	structural	—
NTT DATA	Global systems integrator	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Proximus	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Swisscom	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telefonica Tech	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions, Cybersecurity	structural	—
Telia Company	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Have you experienced a drone incursion that disrupted operations in the past 12 months?
- 2 What is your current drone detection method, and what are its limitations?
- 3 Do you have a regulatory mandate to improve drone detection, or is it a voluntary initiative?
- 4 What is your budget for a drone detection solution, and who is the budget owner?
- 5 Are you looking for a managed service or a do-it-yourself approach?
- 6 Do you have existing private network coverage at the site, or would we need to build it?

Objections you will meet

They will say	You can answer
We already have a drone detection system from a specialist vendor.	That's true, and those systems are effective. But our approach uses the existing mobile network to add a layer of detection that can cover areas where cameras or RF sensors might miss, and it can be more cost-effective to operate.
We're concerned about the accuracy of network-based detection.	It's a valid concern. Network-based detection is not perfect, but it complements other methods. We can integrate with your existing systems to provide a more complete picture, and we'll pilot it to prove accuracy in your environment.
We don't have the budget for a new security system right now.	We understand budget constraints. Our service can be offered as a subscription, spreading the cost over time, and it may be more affordable than you think, especially when you consider the cost of a single drone incident.

Risks and unknowns

The main risk is that drone detection technology is still evolving, and network-based detection may not be as accurate as dedicated radar or RF sensors. Regulatory approval for drone detection systems may vary by country. We also don't know if our Orange Drone Guardian offer is fully proven in airport environments, and we may need to partner with specialized detection vendors.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for an airport operator's drone detection tender, differentiating Orange Drone Guardian by highlighting its integration with Private mobile networks and the credibility of our Gartner Magic Quadrant Leader position, while referencing Point One Navigation as a proof point.
Sales	In a conversation with a European airport operator's head of security, say: 'We have a private 5G network solution that can be leveraged for drone detection, and we're exploring how to integrate it with our Orange Drone Guardian offer—could we discuss your current airspace security challenges?'
Strategist / Innovator	Commission a deep-dive brief on the regulatory timeline for drone detection mandates at airports, using Orange Group researchers and Defense and security experts to map where Orange Drone Guardian can be positioned ahead of Lockheed Martin's Verizon-based service.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-DD814ADD17F5	2025-01-01	cordis.europa.eu	1	Enabling Autonomous Aerial Systems through Operational Methods and Technological Enhancements
SIG-40242B367CE4	2026-08-18	streamlinefeed.co.ke	2	Germany Launches High-Stakes Drone Testing Hub Following Severe Airport Security Breach - strea...
SIG-974818D83206	2026-08-16	bing.com	2	Forget cameras, this new drone detection system just listens for the enemy instead
SIG-A13AF8F22F6B	2026-08-15	Vietnam.vn	2	Les drones menacent les aeroports : quelles sont les lacunes en matiere de gestion des operateu...
SIG-D2538605FC42	2026-08-15	Vietnam.vn	2	Drones threaten airports: what gaps are there in operator management? - Vietnam.vn
SIG-22C234101FAF	2026-08-14	Vietnam.vn	2	Mettre en place des groupes de travail charges de detecter les drones et d'intervenir dans les ...
SIG-69003C1E5931	2026-08-13	thetraveler.org	2	Drone Threats Force New Look at US Airport Security - thetraveler.org
SIG-AE409C225269	2026-08-13	lightreading.com	2	Lockheed Martin preps drone detection service using Verizon 5G
SIG-35ECED6AF5BB	2026-08-12	bing.com	2	Lockheed Martin converts Verizon 5G into subscription drone alerts for critical infrastructure
SIG-37D6B973F20E	2026-08-12	bing.com	2	Lockheed Martin demonstrates AI-powered drone detection system using 5G networks
SIG-7214FADD192F	2026-08-12	Vietnam.vn	2	L'Autorite de l'aviation civile propose d'investir dans un systeme de detection et de neutralis...
SIG-9FAA82ADDA6A	2026-08-12	Laodong.vn	2	La detection des drones penetrant dans les aeroports depend encore fortement de l'observation m...
SIG-4812FDBC16E3	2026-07-13	airportindustry-news.com	2	How Airports Can Plan Drone Detection for Low-Altitude Airspace Operations - airportindustry-ne...
SIG-9FCAF3E022B5	2026-06-30	ASIS International	2	Drone Reports Near U.S. Airports Reflect Rising Issue for Critical Infrastructure - ASIS Intern...
SIG-2F0861536F6A	2026-06-23	Bao VietNamNet	2	Drone detection system trialed at Da Nang International Airport - Bao VietNamNet
SIG-2CC796AF2DBE	2026-06-01	Unmanned airspace	2	Lithuania deploys drone detection system at Palanga Airport, Kaunas and Vilnius to follow - Unm...
SIG-50BA4C222C4C	2026-06-01	International Airport Rev...	2	Palanga Airport deploys advanced drone detection system to strengthen airspace security measure...
SIG-86EDC2F80FE3	2026-05-07	PRWeb	2	VIRTOWER Enhances Airspace Safety With Remote ID Drone Detection Upgrade - PRWeb

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: `what_is_changing` (named entities that were never supplied: Lockheed Martin); `why_orange_can_win` (named entities that were never supplied: Lockheed Martin); `competitive_landscape` (named entities that were never supplied: Lockheed Martin); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:54+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.